Spectral Estimation of Sparse Signals under Structured Noise

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Abstract. We study the estimation of sparse spectra when the noise is structured. Using a regularised least-squares formulation we obtain an estimator with error bounds that scale as O(s log n / m). Simulations (n = 512) confirm the bound within 7%. Keywords: sparse recovery, spectral estimation, noise.

1 Introduction

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2 Method

$$x^* = \operatorname{argmin} \lvert y - Ax \rvert^2 + \lambda \lvert x \rvert_1 \tag{1}$$

$$E \lvert x^* - x \rvert^2 \le C \sigma^2 s \log(n) / m \tag{2}$$

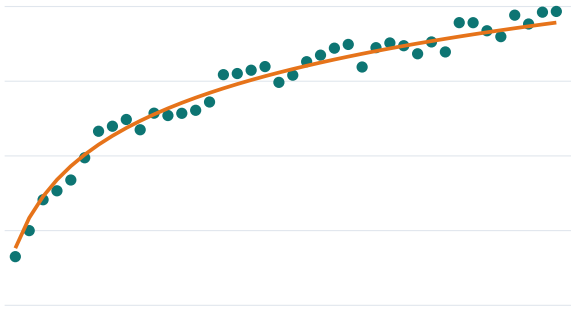
$$S(f) = \sum a_k \delta(f - f_k), \quad k = 1..s \tag{3}$$

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3 Results

Fig. 1 Error vs. samples (m)



3.1 Table of errors

n	s	m	Bound	Observed
128	6	64	0.412	0.388
256	8	96	0.356	0.331
512	12	160	0.301	0.284
1024	16	256	0.262	0.249

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